On Student's 1908 Article "The Probable Error of a Mean"

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This month marks the 100th anniversary of the appearance of William Sealey Gosset's celebrated article, "The Probable Error of a Mean" (Student 1908a). Gosset's elegant result represented the first in a series of exact, "small-sample" results that were developed by Gosset, Fisher, and others to form a central component of the modern theory of statistical inference. This review celebrates the centenary of Gosset's article by discussing both its background and its impact on statistical theory and practice.

KEY WORDS: Cushny and Peebles experiment; Egon Pearson; Fiducial inference; Friedrich Robert Helmert; Physical randomization; Robustness; Ronald Aylmer Fisher; "Student"; t-distribution; William Sealy Gosset.

1. BACKGROUND

William Sealy Gosset (born June 13, 1876, Canterbury, U.K.; died October 16, 1937, Beaconsfield, U.K.) studied at Winchester College and New College, Oxford, U.K. After being awarded a First in the Mathematical Moderations examination in 1897 and a First-Class Degree in Chemistry in 1899, Gossett was employed by Arthur Guinness, Son & Co., Ltd. in Dublin, Ireland in 1899. Gosset recognized almost immediately the importance of statistical methods in the company's operations, and the problem of small samples in particular. "The circumstances of brewing work, with its variable materials and susceptibility to temperature change and necessarily short series of experiments, are all such as to show up most rapidly the limitations of large sample theory and emphasize the necessity for a correct method of treating small samples. It was thus no accident, but the circumstances of his work, that directed 'Student's' attention to this problem" (McMullen 1939, pp. 205–206). At Guinness, Gosset found an environment that encouraged both practical and theoretical research, as Joan Fisher Box (R. A. Fisher's daughter), describes in her informative account (Box 1987, pp. 46–50).

Gosset's "attention" resulted in a report, "The Application of the 'Law of Error' to the work of the Brewery" dated November 3, 1904. (The report, an internal Guinness document, has never been published. Egon Pearson, while preparing his obituary of Gosset, was given "permission [by Guinness] to see and quote from [the report] and other records available in their Dublin brewery"; see Pearson 1939, p. 213.)

Unable to find the necessary results in the literature, Gosset subsequently arranged to meet Karl Pearson during the latter's vacation in July 1905. The meeting was an important one for Gosset; Guinness had an enlightened policy of permitting technical staff leave for study, and a year later Gosset spent the first two terms of the 1906–1907 academic year in Pearson's Biometric Laboratory at University College London.

Thus it was natural for Gosset to consider the question that he did, and equally natural for him to publish it in *Biometrika*, the journal that Pearson co-founded and edited.

2. "THE PROBABLE ERROR OF A MEAN"

Gosset's article has many interesting features, the first of which arises in its second line: its listing of the author.

2.1 "Student"?

In all, Gosset published 14 papers in *Biometrika* (and 7 others elsewhere) over a 25-year period. All but one appeared under the pseudonym "Student." Why "Student"? The solution to this mystery was likely first revealed in the *Journal of the American Statistical Association* more than three-quarters of a century ago by Harold Hotelling (1930, p. 189) in an article on British statistics:

> American students of statistics have long speculated as to the identity of "Student."... I have heard guesses in this country identifying "Student" with Egon S. Pearson and with the Prince of Wales. He is now so well known in Great Britain that no confidence is violated in revealing that he is W. S. Gosset, a research chemist employed by a large Dublin brewery [note that Guinness is not identified by name]. This concern years ago adopted a rule forbidding its chemists to publish their findings. Gosset pleaded that his mathematical and philosophical conclusions were of no possible practical use to competing brewers, and finally was allowed to publish them, but under a pseudonym, to avoid difficulties with the rest of the staff.

Note that Hotelling's account is quite different than the oft-repeated claim that anonymity was demanded by Guinness so that Guinness's competitors would not realize that Guinness found it useful to employ a statistician. According to Hotelling, the anonymity was designed to hide Gosset's identity because of the brewers *inside* Guinness, not those outside!

The pseudonym "Student" was selected by Christopher Digges La Touche, the Managing Director of Guinness: "It was decided by La Touche that such publication might be made without the brewers' names appearing. They would be merely designated 'Pupil' or 'Student' " (Box 1987, p. 46).

Student (as we shall refer to him from now on) was not the only person to benefit from Guinness's enlightened attitude toward statistical education. Guinness sent Edward Somerfield to work with Fisher at Rothamsted in 1922 and George Story to work with Pearson at University College London in 1928. Both were permitted to publish papers in the outside literature, but once again only under pseudonyms: "Mathetes" (Somerfield) and "Sophister" (Story). This subsequent use of pseudonyms indeed may have been motivated by a desire to maintain a competitive edge. Guinness made extensive use of the t-test after its discovery by Student, but little use was made of it elsewhere (McMullen 1939, p. 206). By the 1920s, there was a large statistical group at Guinness, despite the fact that generally "very few scientists in industry made use of mathematical statistical methods [at that time]" (Pearson 1990, p. 91). So it is quite plausible, as Pearson suggests, that Guinness "probably" insisted on

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anonymous publication because it "did not wish it to be known by rival brewers that they were training some of their scientific staff in statistical theory and its application."

2.2 Statistical Philosophy

Student was initially a Bayesian in philosophical outlook, a view he is likely to have gotten from reading Karl Pearson, rather than textbooks on the theory of errors by Airy (1861) or Merriman (1884), Student's early view of the inferential process is very clearly stated in his correlation coefficient paper (Student 1908b), published in the same year as his t-distribution article:

> A random sample has been obtained from an indefinitely large population and r [the sample correlation coefficient] calculated between two variable characters of the individuals composing the sample. We require the probability that R for the population from which the sample is drawn shall lie between any given limits. It is clear that in order to solve this problem we must know two things: (1) the distribution of values of r derived from samples of a population which has a given R, and (2) the *à priori* probability that R for the population lies between any given limits.

But even at this stage, Student made clear his discomfort with the assignment of a prior, adding that:

> Now (2) can hardly ever be known, so that some arbitrary assumption must in general be made; when we know (1) it will be time enough to discuss what will be the best assumption to make, but meanwhile I may suggest two more or less obvious distributions. The first is that any value is equally likely between $+1$ and -1, and the second that the probability that x is the value is proportional to $1 - x^2$. This I think is more in accordance with ordinary experience; the distribution of *à priori* distribution would then be expressed by the equation $y = \frac{3}{4}(1 - x^2)$.

In 1915 and 1917, one can still find Student—in correspondence with Fisher and Pearson—suggesting the use of the priors mentioned in his 1908 article on correlation (see Pearson 1990, pp. 25, 26, 28). But these appear to have been merely casual suggestions; in his letter to Pearson, Student also noted the "disadvantage of using actual knowledge concerning similar work," and in a later letter to Fisher on April 3, 1922, put the case even more strongly:

> When I was in [Pearson's] lab in 1907 I tried to work out variants of Bayes with a priori probabilities other than [a flat one] but I soon convinced myself that with ordinary sized samples one's a priori hypothesis made a fool of the actual sample... and since then have refused to use any other hypothesis than the one which leads to your likelihood [i.e., the flat prior] (where I could deal with the mathematics). Then each piece of evidence can be considered on its own merits (Pearson 1990, pp. 28–29).

Despite this eventual distrust of nonuniform priors (in contrast to Karl Pearson), in his article on the t-distribution, Student's view of the inferential process is clearly Bayesian. He refers to the "usual method of determining the probability that the mean of the population lies within a given distance of the mean of the sample" (p. 1), and throughout his examples refers to the odds that one hypothesis holds rather than another: "The chance that the mean of the population of which these experiments are a sample is positive" (p. 20); "the odds are about 666 to 1 that [one of two drugs] is the better soporific" (p. 21); "the odds are about 14 : 1 that kiln-dried corn gives the higher yield" (p. 24).

One particularly interesting remark is Student's statement that "if two observations have been made and we have no other information, it is an even chance that the mean of the (normal) population will lie between them" (Student 1908a, p. 13). This is in effect an interval estimate; Fisher considered it an early example of a fiducial probability (see Fisher 1939, p. 4; Welch 1958, pp. 782–784).

2.3 Derivation of the t-Distribution

Let us turn to the technical content of Gosset's article. Gosset did not give a rigorous mathematical demonstration of his result. Important elements in the proof already existed (buried in the German literature), but the specific question he asked and answered was both new and novel, and his article inspired much later work by demonstrating the feasibility of adjustment for small sample sizes.

Let $X_1, \ldots, X_n$ denote a sequence of independent and identically distributed random variables, $\mu = EX_j$, $\sigma^2 = \operatorname{var} X_j$, and $X_j \sim \mathrm{N}(\mu, \sigma^2)$. Let $\bar{X}$ and S^2 denote the sample mean and sample variance of the X_j. Then the statistic

$$t := \sqrt{n}\frac{\bar{X} - \mu}{S}$$

has *Student's t-distribution on n* $- 1$ *degrees of freedom*; that is, it has the density function

$$\frac{\Gamma(\frac{n}{2})}{\sqrt{(n-1)\pi}\,\Gamma(\frac{n-1}{2})}\left(1 + \frac{x^2}{n-1}\right)^{-n/2}.$$

The usual derivation of the t-distribution proceeds by showing in some order that

- $\bar{X} \sim \mathrm{N}(\mu, \sigma^2/n)$
- $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$
- $\bar{X}$ and S^2 are independent
- $t \sim t_{n-1}$.

The first of these was well known at the time that Student wrote and could be found in then-standard textbooks on the theory of errors (e.g., Airy 1861, part II, sec. 6, article 43, p. 33; Czuber 1891, pp. 145–147). The first, second, and third sections of Student's article are devoted to deriving successively the last three properties.

Note that, strictly speaking, Student considered the quantity

$$z = (\bar{X} - \mu)\bigg/\sqrt{\frac{\sum_{j=1}^{n}(X_j - \bar{X})^2}{n}}.$$

The transformation $t = z\sqrt{n-1}$ was introduced by Fisher (1925) and later adopted by Student himself (see Eisenhart 1979).

2.3.1 Distribution of S^2. At the time that Student wrote, the distribution of S^2 was known, but not well known (at least in England); it had been derived some three decades earlier in 1876 by the German geodetic scientist Friedrich Robert Helmert (1843–1917). Helmert had a distinguished career, eventually becoming a professor at the University of Berlin and heading its Geodetic Institute (see Fischer 1973). Helmert, following directly in the tradition of Gauss, had earlier published a textbook on the theory of errors (Helmert 1872); thus his interest in the distribution of S^2 was not unnatural. Helmert's textbook became a classic in the field, remaining in print in second (1907) and third (1924) editions until well into the twentieth century. (For detailed discussion of Helmert's derivation, see Hald 1998, pp. 633–637; 2007, pp. 149–152; Sheynin 1995.)

So Student could have just cited Helmert's article for the distribution of S^2. Being unaware of it, however, he instead proceeded to compute the first four moments of S^2 and, noting that, suitably scaled, these agreed with those of chi-squared (a member of the four-parameter Pearson family), correctly conjectured that $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$. To quote Student (p. 5), "hence it is probable that the curve found represents the theoretical distribution of s^2; so that although we have no actual proof, we shall assume it to do so in what follows."

2.3.2 Joint Distribution of $\bar{X}$ and S^2. Let x_j denote the observed value of X_j. Helmert's proof begins by making the change of variable $e_j = x_j - \bar{x}$ and computing the joint density

$$f(e_1, \ldots, e_{n-1}, \bar{x}) = \frac{n}{\sqrt{2\pi\sigma^2}^n} \exp\left\{-\frac{\sum_{j=1}^n e_j^2 + n\bar{x}^2}{2\sigma^2}\right\}$$

(see Uspensky 1937, pp. 333–336 for a proof). This in turn immediately implies the independence of $\bar{X}$ and S^2. Lacking this fact, Student instead showed that $\bar{X}$ and S^2 were uncorrelated and then proceeded as if they were in fact independent. Fisher (1939, pp. 3–4) considered this the most serious gap in Student's argument:

A second theoretical point was treated even more brusquely. The sampling distribution of s is in reality completely independent of that of $\bar{x}$. In order to derive the distribution of the significance test

$$t = \frac{\bar{x} - \mu}{s\sqrt{n}},$$

where μ is the true mean of the population sampled, it was necessary that this independence should be established. "Student" perceived this necessity, and any capable analyst could have shown him the demonstration he needed. As it was, he satisfied himself with showing somewhat laboriously that the distribution of s^2 was uncorrelated with both $\bar{x}$ and with $\bar{x}^2$. This was the most striking gap in his argument, for in truth it was not merely the distribution of s found by Helmert, but the exact simultaneous distribution of s and $\bar{x}$ that "Student" needed to develop his test.

Suppose that the summands X_j have any distribution with a third moment. If $\mu_3 = E[(X_j - \mu)^3]$, then

$$\text{cov}(\bar{X}, S^2) = \frac{\mu_3}{n}$$

(see, e.g., Zhang 2007, pp. 159–160). This simple equation sheds light on Student's observation; the fact that $\bar{X}$ and S^2 are uncorrelated simply reflects the absence of skewness in the case of the normal, and indeed holds whenever μ_3 vanishes. The *independence* of $\bar{X}$ and S^2, in contrast, is much more restrictive; it is both a necessary *and* sufficient condition for normality, as first noted by Geary (1936) (also see Lukacs 1942).

Deriving the joint distribution of $\bar{X}$ and S^2 has had a recurrent fascination for statisticians since the work of Fisher (1915). Kruskal (1946), for example, found an interesting inductive proof inspired by an earlier computation of Helmert; this in turn inspired Stigler (1984) to produce an elegant reformulation of Kruskal's proof. (For discussion of Stigler's proof, see Zehna 1991 and the ensuing debate in the "Letters" section of *The American Statistician*, 1992, vol. 46, pp. 70–77.)

2.3.3 Distribution of t. Given these results, it was a straightforward exercise in integration for Student to derive the distribution of t. Perversely, here too he had been (partly) anticipated, by Jakob Lüroth (1876) and Francis Ysidro Edgeworth (1883), both of whom had derived the t-distribution as a posterior distribution for the population mean using a flat prior, Lüroth for the more general case of linear regression and Edgeworth in the one-sample case considered by Student (Welch 1958, p. 779).

It is not entirely surprising that Student (as well as *Biometrika* and its readership) were unfamiliar with Helmert's work. In general, there was a considerable gulf between continental and English statistics both at the time that Student wrote and for some time after. The reasons for and extent of this divide could easily be the subject of a lengthy article in itself.

Karl Pearson (the editor of *Biometrika*) eventually did note (in 1931!) Helmert's priority in an editorial in *Biometrika* (Pearson 1931, p. 416):

The familiar distribution of the standard deviations of samples from an indefinitely large normal population appears to have been discovered on several occasions—without doubt independently—and by later writers attributed to various investigators. There can, we think, be no doubt that the original discoverer was Helmert and that he much antidates other claimants.

What was the reason for such persistent neglect? Pearson could give no good explanation, noting that Helmert's article had appeared "in a journal which would be rarely consulted by statisticians," but conceding at the same time that "Helmert's result has been reproduced in German textbooks on the theory of observations" (e.g., Czuber 1891). Pearson suggested that the formula be called "Helmert's equation."

2.4 "Practical Test of the Foregoing Equation"

One of the most interesting aspects of Student's article is his use of simulation to investigate the small-sample behavior of z $(= t/\sqrt{n-1})$. He noted the following:

Before I had succeeded in solving my problem analytically, I had endeavoured to do so empirically. The material used was a correlation table containing the height and left middle finger measurements of 3,000 criminals, from a paper by W. R. Macdonell (*Biometrika*, vol. I, p. 219). The measurements were written out on 3,000 pieces of cardboard, which were then very thoroughly shuffled and drawn at random. As each card was drawn its numbers were written down in a book which thus contains the measurements of 3,000 criminals in a random order. Finally each consecutive set of 4 was taken as a sample—750 in all—and the mean, standard deviation, and correlation of each sample determined. The difference between the mean of each sample and the mean of the population was then divided by the standard deviation of the sample, giving us the z of Section III.

It has sometimes been suggested that this represents the earliest instance of the use of simulation in statistics (see, e.g., Pearson 1939, p. 223; Teichrow 1965); but Stigler (1991) pointed to several earlier examples in the nineteenth century (apart from Buffon's classic needle experiment in the eighteenth). Nevertheless, it is certain that Student's simulation is one of a small number performed before the appearance of L. H. C. Tippett's table of random numbers in 1927. After Tippett (1925, p. 378) and Church (1926, pp. 324–325, 332–333) had encountered difficulties in carrying out random sampling experiments for simulation studies reported in *Biometrika*, Karl Pearson suggested that Tippett prepare a table of random numbers to facilitate such experiments. The result was Tippett's *Tracts for Computers* no. XV, published in 1927 (see Pearson 1990, pp. 88–95). Note that the term "computer" in Tippett's title refers to *human* computers, a usage that continued through World War II.

It is a pity Student does not expand on how his cards were "very thoroughly shuffled." Although his results appear to have been satisfactory, true physical randomization is often very difficult to achieve, and in the hands of the inexperienced often is not achieved. The 1970 draft lottery debacle is a celebrated instance (see Fienberg 1971). Rowlett (1998, pp. 53–54) provided an amusing description of some of the difficulties that U.S. codemakers encountered during the 1930s when they tried to mix thousands of cards for a codebook.

2.5 "Illustrations of Method"

Technometrics states in its "Information for Authors" that "every article shall include adequate justification of the application of the technique, preferably by means of an actual application to a problem in the physical, chemical, or engineering sciences." Student, given his strong practical bent, needed no such directive; his article contains no fewer than four applications to real data. (It should be noted that all of these involve one-sample t-tests.)

The first of Student's four examples concerned a famous experiment investigating whether different optical isomers might have different sleep-inducing effects. Student (1908a, p. 20) noted that:

> As an instance of the kind of use which may be made of the tables, I take the following figures from a table by A. R. Cushny and A. R. Peebles in the *Journal of Physiology* for 1904 [sic], showing the different effects of the optical isomers of hyoscyamine hydrobromide in producing sleep. The sleep of 10 patients was measured without hypnotic and after treatment (1) with D. hyoscyamine hydrobromide, (2) with L. hyoscyamine hydrobromide. The average number of hours' sleep gained by the use of the drug is tabulated below.

This is a celebrated data set (Cushny and Peebles 1905) in part because Fisher used Student's tabular presentation of the data as an example in his classic textbook *Statistical Methods for Research Workers* (1925; hereinafter referred to as SMRW, sec. 24). Student stated that the drugs in question were the levo and dextro forms of hyoscyamine hydrobromide, a statement accepted by Fisher in the earlier editions of his book. However, Dr. Isidor Greenwald of the New York University and Bellevue Hospital Medical College pointed out to Fisher in a letter (dated December 12, 1934) that the two drugs corresponding to the data in question were L-hyoscyamine hydrobromate and L-hyoscine hydrobromate—not optical isomers at all! Fisher—who disliked major surgery in his books—dealt with this unpleasantness by replacing the names of the drugs in the two columns by "A" and "B" in later editions of SMRW! The episode is a useful cautionary tale in the use of secondary sources for information and data, especially when the original data are readily available. (The correspondence was given in full in McMullen 1970, and extracts were quoted in Pearson 1990, p. 54.)

But the problems with Student's (and Fisher's) analysis extend far beyond mere mislabeling. The "A" and "B" measurements are averages based on samples varying in size from 3 to 6, so the measurements have no σ in common for S to estimate! (For detailed discussion of both the original Cushny and Peebles experiment and data and Student's analysis of it, see Preece 1982; Senn and Richardson 1994; Senn 2002, pp. 1–7.)

2.6 Robustness of the t-Distribution

Of course, no real distribution is exactly normal, and so the practical applicability of the t-distribution must depend on its relative *robustness* to departures from normality. What is needed, furthermore, is not merely an asymptotic result, but some connection to our everyday experience. Some forms of limiting behavior can take a very long time to set in. The central limit theorem vindicates its name in part because normality sets in both early and often.

Student was sensitive to this issue. In his Introduction, he concedes that his conclusions "are not strictly applicable to populations known not to be normally distributed," but asserts that "it appears probable that the deviation from normality must be very extreme to lead to serious error." This claim was later supported by empirical studies in Section VI (p. 19): "I believe that the tables at the end of the present paper may be used in estimating the degree of certainty arrived at by the mean of a few experiments, in the case of most laboratory or biological work where the distributions are as a rule of a 'cocked hat' type and so sufficiently nearly normal."

The (surprising?) robustness of the t has been perhaps one of its greatest practical features to emerge in the years since 1908. There have been many studies of the distribution of the t-statistic under departures from normality (one example in *JASA* is Efron 1969). It is interesting that one natural question, when the t-statistic is asymptotically normal as $n \to \infty$, remained open for nearly a quarter of century after its answer was suggested by Logan, Mallows, Rice, and Shepp (1973). Giné, Götze, and Mason (1996) were ultimately able to prove that the necessary and sufficient condition on the summands X_j of the t-statistic to ensure convergence to the normal is that $EX_j = 0$ and X_j lie in the domain of attraction of the normal distribution. Such results appear in the literature as part of the theory of *self-normalized* (or "studentized") sums.

By any measure (citation, theoretical extension, practical use), Student's two 1908 articles initially attracted surprisingly little notice. Egon Pearson (1939, p. 225), who was certainly in a position to judge, observed that "one of the curious things that must strike us now about these two papers of Gosset's... is the small influence that their publication had for a number of years on current statistical literature and practice." But eventually Student "acquired a new champion of exceptional brilliance and enormous energy" (Lehmann 1999, p. 419).

3. RONALD AYLMER FISHER

When Isidor Greenwald wrote Fisher in 1934 about the Cushny–Peebles gaffe in SMRW, he concluded by remarking "I am greatly surprised that this error should not have been corrected long ago" (Pearson 1990, p. 54). When Student wrote back to Fisher a week later acknowledging the error, he remarked in a postscript that "of course it is not surprising that no one discovered the blunder for in the pre Fisher days no one paid the slightest attention to the paper." The history of Student and the history of Fisher are inextricably linked. Fisher not only championed Student's work, but Student exerted a profound influence on the nature and direction of Fisher's research for nearly two decades.

3.1 Fisher Meets Student

Fisher's supervisor at Gonville and Caius (Fisher's college at Cambridge), the astronomer Frederick J. M. Stratton, had initially put Fisher in touch with Student, suggesting he send Student a copy of his first paper (Fisher 1912). Mathematicians, especially young and ambitious ones fresh out of school, are always looking out for crisply defined problems on which they can bring their prowess to bear. R. A. Fisher was no exception. Providing a rigorous mathematical derivation of the t-distribution was a natural challenge, and in 1912 Fisher wrote to Student, giving a complete and rigorous derivation of the t-distribution, exploiting his facility in n-dimensional geometry. Student forwarded the proof to Pearson, suggesting that the result might be the subject of a note in *Biometrika*, but Pearson demurred, responding that he was unable to follow Fisher's proof (Pearson 1968, pp. 446–447; 1990, pp. 47–48). Fisher likely never learned this; in his obituary notice of Student, Fisher (1939, p. 5) stated that "in his correspondence with me [Student] did not even suggest that the completed proofs should be published."

This was merely a temporary setback. Fisher was later able to find a geometric proof for the distribution of the correlation coefficient as well, and the resulting article published in *Biometrika* was his first important contribution to statistics. Fisher (1915, p. 507) wrote, referring to Student's derivation of the t:

This result, although arrived at by empirical methods, was established almost beyond reasonable doubt in the first of "Student's" papers. It is, however, of interest to notice that the form establishes itself instantly, when the distribution of the sample is viewed geometrically.

The formal proof itself was given later by Fisher (1923).

3.2 Ubiquity of the t

This was only the tip of the iceberg. On April 3, 1922, Student wrote to Fisher asking about the distribution of a regression coefficient, and followed this up on April 9 with a similar question about partial correlation and regression coefficients. Fisher replied by early May that the t-distribution supplied the answer to both of Student's questions as well as the case of the difference of two means (McMullen 1970, letters 5–7; Pearson 1990, pp. 48–49; Box 1981, pp. 62–63). These results were reported in part by Fisher (1922), summarized in his article "Applications of 'Student's' Distribution" (Fisher 1925), formed the basis of his address to the International Congress of Mathematicians at Toronto in 1924 (Fisher 1924), and set out for practical use in *Statistical Methods for Research Workers*.

This explosion in the potential uses of the t-distribution led to new tables, published jointly by Fisher and Student in *Metron*, based in part on new expansions derived by Fisher (see Eisenhart 1979; Box 1981, pp. 63–64). It was precisely at this point that the passage from Student's initial use of "z" to Fisher's "t" took place.

3.3 Robustness Revisited

Student, as we have seen, was certainly alert to the issue of the sensitivity of his approach to possible departures from normality. Clearly, such concerns would apply with equal force to Fisher's methods, vigorously championed in SMRW.

3.3.1 Egon Pearson's Review of SMRW. When the second edition of SMRW appeared in 1928, Egon Pearson raised this issue in a review in *Nature*, angering Fisher. Pearson (1929, pp. 866–867) wrote:

There is one criticism, however, which must be made from the statistical point of view. A large number of tests are developed upon the assumption that the population sampled is of normal form. That this is the case may be gathered from a very careful reading of the text, but the point is not sufficiently emphasised. It does not appear reasonable to lay stress on the exactness of tests, when no means whatever are given of appreciating how rapidly they become inexact as the population samples diverges from normality. That the tests, for example, connected with the analysis of variance are far more dependent on normality than those involving "Student's" z (or t) distribution is almost certain, but no clear indication of the need for caution in their application is given to the worker. It would seem wiser in the long run, even in a text-book, to admit the incompleteness of the theory in this direction, rather than risk giving the reader the impression that the solution of all his problems have been achieved. The author's contributions to the development of normal theory will stand by themselves, both for their direct practical values and as an important preliminary to the wider extension of theory, without any suggestion of undue completeness.

Ironically, Pearson's concerns had been strongly influenced by Student. Student had written to Pearson on May 11, 1926, raising the issue of the robustness of the rapidly burgeoning number of tests based on normal distribution theory. The existence of Tippett's tables made possible a systematic study of the issue, a study that Pearson proceeded to carry out between 1928 and 1931 (Pearson 1990, sec. 6.4).

Student acted as a mediator between Fisher and Pearson after Pearson's (unsigned) review appeared in *Nature* on June 8, 1939; in the end, *Nature* published letters responding to the review by both Student (on July 20) and Fisher (on August 3) (see Pearson 1990, sec. 6.5). One comment from Fisher is of particular interest because of the reaction that it elicited from Student. Fisher wrote:

I have never known difficulty to arise in biological work from imperfect normality of the variation, often though I have examined data for this particular cause of difficulty; nor is there, I believe, any case to the contrary in the literature.

Student made the following comment to Pearson in a letter on September 25 (Pearson 1990, p. 99):

Fisher is only talking through his hat when he talks of his experience; it isn't so very extensive and I bet he hasn't often put the matter to the test; how could he?

How ironic coming from the person who in so many ways inspired Fisher's development of statistical methods based on the normal distribution!

3.3.2 Fisher's Randomization Test. The third chapter of Fisher's classic book *The Design of Experiments* (Fisher 1935a) is devoted to an analysis of Charles Darwin's *Zea mays* data. In brief, these data comprise the heights of 15 pairs of plants, one member of the pair being a cross-fertilized plant, the other a self-fertilized plant. In Section 17 ("Student's" t test) the data are summarized in Table 3 as 15 differences in eighths of an inch, and used to illustrate use of the t-test.

Later on, in Section 21, noting that some statisticians had stressed the assumption of normality in the hypothesis being tested, Fisher advanced an early example of a nonparametric test:

[T]he physical act of randomisation, which, as has been shown, is necessary for the validity of any test of significance, afford the means, in respect of any particular body of data, of examining the wider hypothesis in which no normality of distribution is implied.

Suppose that there are n paired differences (here $n = 15$). Let x_i be the height of a cross-fertilized plant, y_i be the height of a self-fertilized plant, $d_i = |x_i - y_i|$, and

$$S_n = \epsilon_1 d_1 + \cdots + \epsilon_n d_n$$

for some choice of $\epsilon_i = \pm 1$. Fisher's (typically clever) idea was to view the observed value of S_n as one of 2^n possible such values, and to see whether the observed value was unusual compared with the other possible values. Formally, this can be considered a test of the null that the observed value S_n has been randomly selected from the set of all possible sums.

In the case of Darwin's data, $n = 15$, $2^n = 32{,}728$, the observed value of S_n is 314, and there are 1,726 possible values of S_n such that $|S_n| > 314$; thus the level of significance for the randomization test is $1{,}726/32{,}728 = .05267$. In contrast, the level of significance computed based on the t-test is .0497. The p value for the t-test turns out to be an excellent approximation to the p value for the randomization test! Diaconis and Holmes

(1994) explored the use of Gray codes to expedite the computation of randomization distributions and discuss Fisher's *Zea mays* example as an illustration; their figure 2.1 shows that the randomization distribution for this set of data is bell-shaped (indeed remarkably so), and their figure 2.2 shows that both the normal and t provide excellent approximations to the randomization distribution.

3.4 Fiducial Inference

The t-distribution has served as a touchstone for fiducial inference throughout its history. Fisher, in *The Design of Experiments* (1935a, sec. 62), credited the biologist E. J. Maskell (who worked at Rothamsted in the early 1920s) with the observation that one could use the percentiles of the t-distribution to give interval estimates for μ corresponding to any desired level of significance. Such use eventually morphed into Fisher's concept of a *fiducial distribution*, although Maskell's precise role in its birth is a matter of dispute (compare Edwards 1995, pp. 800–801 and Aldrich 2000, p. 158). In any event, Fisher regarded the use of the t a pivotal quantity to be a basic example of the fiducial argument (see Fisher 1935b, pp. 391–394).

Fisher revisited this use of the t in his final book, *Statistical Methods and Scientific Inference* (Fisher 1956, chap. 4, sec. 2). In the intervening two decades, however, his view of the fiducial argument had radically changed; his defense of fiducial inference now centered around the idea of a *recognizable subset*. In the case of the t, for example, Fisher claimed that in considering the inequality

$$\mu < \bar{x} - \frac{1}{\sqrt{N}} ts,$$

"there is no possibility of recognizing any sub-set of cases, within the general set, for which any different value of the probability should hold" (ibid., p. 84). No mathematical proof of this assertion was given, however, and Buehler and Feddersen (1963)—perhaps somewhat surprisingly—were able to show that recognizable subsets do in fact exist in this case (see also Brown 1967). (For subsequent defences of the fiducial argument addressing this issue, see Yates 1964 and Barnard 1995.)

Jeffreys (1931, 1937, 1939) discussed Bayesian derivation of the t as a posterior distribution for the mean, and explored the reasons why Fisher's derivation of the fiducial distribution of the mean comes to the same thing. Note that Edgeworth (1883), Burnside (1923), and Jeffreys arrived at t-posteriors with different degrees of freedom because they each used (in effect) different priors for σ.

4. SOME LITERATURE

An enormous amount has been written about Student, his work, and his 1908 article. The following is intended as a brief tour through this thicket.

Of the appreciations that came out at the time of Student's death, those of Fisher (1939), McMullen (1939), and Pearson (1939) are of particular interest; the last of these gave a detailed and lengthy overview of the totality of Student's work. Egon Pearson's statistical biography of Student (Pearson 1990), edited by Plackett and Barnard after Pearson's death in 1980, contains a wealth of further information and quotes extensively from correspondence Student had with both Karl and Egon Pearson. Fisher's biography by his daughter Joan Fisher Box (1978) is also useful.

Student's *Collected Papers* (Pearson and Wishart 1942) has long been out of print, but most of his important papers appeared in *Biometrika*, and thus are available online through *JSTOR*. Student's correspondence with Fisher (McMullen 1970) has, unfortunately, appeared only in privately published form. Excerpts from and discussion of the correspondence have been given by Box (1981) and Pearson (1990, chap. 5).

For a modern technical discussion of Student's work, one can consult Hald's comprehensive and meticuluously documented historical studies (Hald 1998, 2007). Welch (1958) provided an interesting view at mid-century of Student's work, published in *JASA* to mark the 50th anniversary of Student's article. It will be interesting to see what *JASA* publishes 50 years from now!

5. CONCLUSION

Student's article is truly remarkable for its richness. It simultaneously heralded the advent of small-sample distributional studies in statistics, used simulation in a serious way to investigate such distributions, and investigated the robustness of its results against modest departures from normality. It was a singular accomplishment, especially so given the limited formal background in mathematics and statistics that Student had. But it took the genius and drive of a Fisher to give Student's work general currency. The contribution of Fisher to our profession is widely appreciated; the extent of Student's contribution perhaps not as much.

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Comment

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Sandy Zabell's excellent account of Student's 1908 article is exemplary in all respects. I will add only a brief comment on the question of the role of Gosset in the historical development of statistics.

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In 1990 I had the occasion to review for ISI's *Short Book Reviews* the volume *"Student": A Statistical Biography of William Sealy Gosset*. The book (Pearson 1990) had been begun by Egon Pearson and was completed after Pearson's death by Robin L. Plackett and George A. Barnard. In the course of the review, I commented on Gosset's admirable character and attractive writing style, but ended on a provocative note:

> Gosset possessed excellent statistical insight, and he was surely a catalyst to some important developments in statistics. But there has long been a tendency to exaggerate his achievements, I suspect in recognition of his admirable character, and without a more extensive study it is difficult to judge whether he was an essential catalyst.

As it happened, Frank Yates had been asked to review the book before it came to me but had declined (after a reading that left marginal notes on the copy that came to me). Although Yates, who had known Gosset, may not have found the book sufficiently to his taste to review, he nonetheless rose to my provocation to write a spirited rejoinder, which the editor of *Short Book Reviews* published in a later issue, the only time in 25 years that such a reply appeared. Yates, not surprisingly, thought Gosset was much more than a catalyst, but I still think my observation was a fair one, and I do not pretend to know even now the answer to the question I asked, of whether he was an *essential* catalyst; that is, would the development of statistics have been much different had Gosset never traveled to study in Pearson's laboratory in London?

Of course, I do not mean to raise any doubt about the excellence of Gosset's work or the quality of his mind; Sandy seems to me to capture those quite accurately. But there are examples of people who have written excellent works and who have later attracted renown, but who arguably had little or no impact on the development of our subject. Thomas Bayes is one such example. His article received no serious notice before the twentieth century, and I think a case might be made that had his article never been written, only our modern terminology would be different.

Gosset's case is different. As Sandy explains, whereas the 1908 article was essentially ignored by most statisticians for more than two decades, there was a key exception: Fisher. The article's effect on Fisher was clearly important; it surely played a role in exciting his interest in problems of distribution. Gosset's technique (essentially Pearsonian moment calculation and curve fitting) had no visible effect on Fisher, who immediately adopted a totally different approach; what influence the article had was due to the problem it addressed, not to Gosset's attempted solution, I would suggest. Even if Gosset's guess of the distribution of s^2 had been wrong, I think the effect would have been the same, and the article gave no hint of any of the magic that Fisher produced in the beautiful interrelations among the distributions of the t-statistic, the two-sample t-statistic, the correlation coefficient, regression coefficients, and the sums of squares for analysis of variance.

Fisher's laudatory obituary of Gosset (Fisher 1939) may be read as supporting this view. After four and a half pages of the highest praise for Gosset (and a few digs at Pearson), Fisher took much of it back by stating that he doubted Gosset had understood the full measure what he had done. Indeed, I think Fisher was accurate in that assessment, if we take his description of Gosset's accomplishment at face value. As Alfred North Whitehead wrote in 1917, "Everything of importance has been said before by someone who did not discover it." From this point of view, the importance of the 1908 article is due to what Fisher found there, not to what Gosset placed there. That may have been self-serving on Fisher's part, but I think there is merit in the view, and the fact that no one else found gold in that vein between 1908 and 1922 argues in its favor.

I still view Gosset's primary role as that of an important catalyst. The question of essentiality may be unanswerable, and certainly has no bearing on the decision to celebrate Gosset's achievement of 1908. Gosset was a wise and creative worker who, although thoroughly in the sway of nineteenth century and Pearsonian ideas, wrote one article that caught the eye of the one person who could break free of those constraints. Sandy Zabell's lucid and scholarly analysis of his life and work gives a perfect accent to the occasion of celebrating that article and the man who will always be best known as "Student."

Comment

John ALDRICH

1. INTRODUCTION

When we think of "Student's t," we are at least as likely to be thinking Ronald Fisher's thoughts as Student's. The designation, "t-distribution with $n-1$ degrees of freedom," like the idea of t as one of a family of distributions based on the normal distribution or the application of the t-distribution to regression, were products of Fisher's imagination. That we think of Student's article at all is due largely to Fisher. Professor Zabell quotes Student, writing in 1934, "in the pre Fisher days no one paid the slightest attention to the paper." I would like to develop the theme of Student and Fisher and the "inextricable link" between their histories.

After 1908, Student wrote only three articles on his z distribution; two extended the original tables, and one replied to a criticism that Karl Pearson made in 1931(!). The big advances were made by Fisher, whose studentry can be divided into two phases; in the first phase; he did or redid Student's mathematics, and in the second phase, he took over the z distribution and reconfigured it. The second phase, one of the great passages

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in the history of twentieth century statistics, was well treated in the biographies by Box (1978, chap. 5) and E. S. Pearson (1990, chap. 5), whereas Eisenhart (1979) followed the transition from the z of Student (1908a) to the t of Fisher (1925a). Of course, the transition went deeper than just writing $t = z\sqrt{n-1}$. The not so familiar first phase was not so spectacular, but is quite intriguing nonetheless. Fisher (1939) wrote his own record of Student's scientific contribution, but his aim was to instruct, and anything that might distract from the lesson, such as Student's Bayesianism or his own reasons for working on Student's problems, was omitted.

The story will also tell us something about Student the person. The Student–Fisher connection is that rare thing, a sunny story from the history of British statistics in the early twentieth century. Much of the sunshine came from Student, and the story clearly brings out the kind of person he was.

2. THE VERY BEGINNING

Fisher and Gosset first met in September 1922, but they had been corresponding intermittently for 10 years. In 1912, when the first phase of their "collaboration" began, Fisher was 22 years old and Gosset was 36; Karl Pearson was 55. This first phase ended in 1915 with the publication of Fisher's article on the exact distribution of the correlation coefficient; this is the only publication from that time to reveal any connection between the two men's work. They already recognized each other's qualities: "It has been the greatest pleasure and interest to myself to observe with what accuracy 'Student's' insight has led him to the right conclusions" Fisher (1915, p. 508) reports. Student thanked Fisher for "the kind way in which you referred to my unscientific efforts" (Pearson 1968, p. 447). Fisher added mathematical precision to Student's insight; Student set up problems, and Fisher knocked them over. Doing the mathematics for Student was not a small thing, for nobody else could. Zabell identifies one factor behind Fisher's taking on this role—the desire of a young and ambitious mathematician to show what he could do; but Fisher already had his own scientific agenda, and Student's problems happened to be on it.

Fisher's 1915 article grew out of Student's second 1908 article on "the probable error of the correlation coefficient." But, as Zabell relates, there was an earlier nonpublished article based on the first in which Fisher derived the distribution for z. In September 1912, Student forwarded Fisher's derivation to Karl Pearson, suggesting that he publish it. In the covering letter (reproduced in Pearson 1968, p. 446), Student summarized his transactions with Fisher. These began a few months earlier when Fisher sent Student an article that he had written (Fisher 1912) that proposed a new estimation method, the "absolute criterion." Fisher later called this method "maximum likelihood." He applied the method to the problem of estimating the mean and the precision of the normal distribution. Fisher's estimate for the precision (parameterized as $h = 1/\sigma\sqrt{2}$) involved a factor n instead of the $(n-1)$ that was customary in the theory of errors. Fisher criticized some arguments leading to the $(n-1)$ value, but then the argument took an unexpected turn. Fisher mentioned that h could be estimated by choosing the value that maximizes the frequency distribution of the statistic that Student (unknown to Fisher) denoted by s^2. This second procedure actually supplanted the first in Fisher's thinking, for when Fisher (1915) used an analogous procedure to estimate the correlation coefficient by maximizing the frequency distribution of r with respect to ρ, he referred to it as the "absolute criterion." Fisher only gave up this second form of the absolute criterion (for the first) in 1921 (for a more complete account, see Aldrich 1997). Fisher had an interest in obtaining the distribution of s^2, but whether or not he derived it independently of Student (1908a) is unclear; that article must have come up in the course of their correspondence. At the time, Fisher had no real business with the z distribution, and Zabell is probably right that Fisher derived it because he could!

Together, Student's articles of 1908 and Fisher's work of 1912–1915 produced a collection of results in distribution theory, and yet, if I am right, their collaboration was based on different priorities and conflicting approaches to inference. Student was interested in producing a test based on z, and for this the distribution of s was just an input; for Fisher, the distribution of s was wanted for estimation and the distribution of z came as an easy extension. Regarding principles of inference, Student was a Bayesian—of a kind. Zabell (Sec. 3.12) describes the curious Bayesian structure of the correlation article in which a frequency distribution for the correlation coefficient was sought so it could be multiplied by a prior, and also how the z frequency distribution was treated as a posterior without any explicit Bayesian structure to support it. Like Student's thinking about z, Fisher's thinking about the absolute criterion was half-baked, yet there is a clear anti-Bayesian streak in his 1912 article. Student and Fisher seem to have converged on the same problems for unrelated, if not opposed, reasons. To what extent they *exchanged* views on inference is unknown, for only one letter survives—Student's thank you letter for the correlation article. This is the letter to which Zabell refers for Student's pondering the effect of adopting different priors.

In 1908, Student sought exact distributions for three quantities, s, z, and r. The publication of "The Probable Error of a Mean" was *not* a great event. Student had taken the problem to Pearson, who had helped him solve it. Student used Pearson's tools and wrote in his language (see Aldrich 2003), but the problem was not Pearson's, and its solution gave him no cause for celebration. The problem belonged to the Gaussian theory of errors, which Pearson considered defunct. Student's tables went into Pearson's *Tables for Statisticians* (1914); good tables were not to be wasted, even if they were good for very little. Otherwise, Pearson ignored Student's z until 1931. For the biometrics of the time, only the distribution of r really mattered, although Fisher, quixotically, became interested in s as well. The distribution of s was brought up by Fisher in his correlation article (1915, p. 509) and in a postscript to that article, Pearson (1915) added his thoughts on s. Buried in Fisher's work (1915, p. 518) was one new use for Student's z, but Fisher's interest in z caught fire only when he found a use for it in regression in 1922. It was then that the t story took off.

3. THE t DISTRIBUTION AND STUDENT

In December 1918 (contact had been reestablished in 1917), Student told Fisher that there might be a job going at Rothamsted Experimental Station: "I don't know whether you are looking for a job in that line, but I hear that Russell intends to get a statistician sometime soon." Guinness had an interest in the cultivation of barley, one of its main inputs, and Student was a figure in the world of agricultural experiments. Fisher was offered the Rothamsted job, and agricultural experiments became his line. E. S. Pearson (1968, p. 448) thought it "very likely" that Fisher's appointment owed something to Gosset's links with the agriculturalists.

In the early years at Rothamsted, Student was Fisher's lifeline to the community of statisticians. He was the first to be told of new results, and Fisher considered him the only person who understood his work. More than 60 letters survive from the period 1922–1925, nearly all from Student to Fisher. A request in the letter of April 3, 1922 (letter 5 in McMullen 1970) precipitated the second phase of Fisher's studentry. Professor Zabell quoted from that letter when describing Student's views on Bayes. Student had seen the first of Fisher's articles on "likelihood" and knew that Fisher rejected the Bayesian argument. He also had seen the first of the articles reconstructing Pearson's chi-squared theory; not only was Fisher discussing significance tests in earnest, but he also had introduced the notion of "degrees of freedom." Gosset wrote: "I want to know what is the frequency distribution of $r\sigma_x/\sigma_y$ for small samples, in my work I want that more than the r distribution now happily solved." From the notation and reference to r, Student was clearly after the solution to another problem associated with the bivariate normal, the distribution of the regression coefficient. Fisher's response surprised him, because it involved relocating the problem from the theory of correlation to the theory of errors; the change was discussed by Aldrich (2005). The solution involved a suitably constructed z statistic, which pleased Student, although he was not easily persuaded that the solution was correct.

A year later, in May 1923, Fisher was reporting an advance of a different kind, an account of the interrelationship between the various distributions associated with the normal distribution. The letter (which appears in Box 1978, p. 118) formed the basis of the synthesis, "On a Distribution Yielding the Error Functions of Several Well-Known Statistics" Fisher (1924), in which Pearson's chi-squared and Student's t distributions (for the first time) appear as special cases of a general distribution that Fisher called z; transformed, this became the modern F $(= e^{2z})$. Amid the general advance, one backward look should be mentioned. The group theorist William Burnside (1923) published a treatment of the Bayesian version of the problem of the probable error of the mean; Pfanzagl and Sheynin (1996) described this work. Fisher wrote a note (1923) registering Student's priority and giving a derivation of z on the lines, presumably, of the rejected piece from 1912. Fisher also provided a clear statement of the difference between the Bayesian and sampling theory projects. When Fisher sent Student Burnside's paper and his own note, Student simply commented, "It is interesting to see how à priori probability has got him just off the line." (letter 25 in McMullen 1970). There was no degrees of freedom adjustment, as it would be called later. In a later letter (letter 39), Student referred to the "futility of à priori assumptions." It is interesting to speculate on how he would have reacted to the argument of Jeffreys (1931), which gave a distribution exactly on the line (i.e., a t with the right number of degrees of freedom); whether he ever saw it is not known.

The new t was proclaimed in two works in 1925. "Applications of 'Student's' Distribution" provides the theory of the applications, and *Statistical Methods for Research Workers* demonstrates the applications. The book would make both Fisher's and Student's names. It is largely a book of three distributions, Fisher's z for the analysis of variance and two from others, Pearson's chi-squared and Student's t. Suddenly the occasional contributor of minor pieces to *Biometrika* was on the pedestal with the master, and, more than that, *his* contribution contained no "serious error." In the fourth edition Fisher (1932, p. 24) wrote that "from the first edition it has been one of the chief purposes of this book to make better known the effect of [Student's] researches, and of the mathematical work consequent upon them."

Fisher also reworked Student's old examples. Zabell (Secs. 3.2 and 3.5) notes how Fisher used one of Student's data sets, the Cushny–Peebles data, to illustrate the t-test. Naturally, Fisher (1925a, p. 108) stripped away the Bayesian language; instead of saying that "the odds are about 666 to 1 that 2 is the better soporific," Fisher concluded from the t value of 4.06 that "only one value in a hundred will exceed 3.250 by chance so the difference between the results is clearly significant." If Student did not like this reformulation, he had at least two opportunities to say so. He read the proofs of the book as a favor to Fisher and reviewed the published work (Student 1926); on neither occasion did he comment on Fisher's handling of Student's distribution. He was not afraid of registering disagreement; he was always skeptical about the use of controlled randomization in experimental design. At the proof stage (letter 50), he commented, "you would want a large lunatic asylum for the operators who are apt to make mistakes enough even at present." He made this point more decorously in the review.

Student's "new tables" of 1925 were for t. Student (1925, p. 105) saw two defects in the existing tables: "as n increases, the z scale becomes very coarse" and "except in the case for which it was designed, n, the number in the sample, is not the best number under which to enter the table, but $n - 1$ the number of degrees of freedom." Student deferred to Fisher in mathematics—in letter 36 he refers to his "Watsoning" to Fisher's Holmes—and he saw Fisher's replacement of z by t as a mathematical advance. But beyond the mathematics, Student's final statement on z is fully Fisherized and entirely de-Bayesed. To correct Karl Pearson's misunderstanding of the z-test (Pearson never acknowledged the existence of t), Student (1931, p. 408) spelled out "what we actually ask ourselves":

> If the average difference between A and B in the population were zero, what would be the probability of obtaining a sample of differences giving a value of z as high as that observed? and if this probability is sufficiently small we say that the difference is significant.

4. FISHER AND STUDENT

In the early years, Fisher received valuable support from Student, the one established statistician who believed in him, and Fisher (1939, p. 8) acknowledged a "loyal and generous friend." Student the man did not need Fisher's help, but to Student the scientist ("one of the most original minds in contemporary science"), Fisher was very generous. Fisher (1939, pp. 5–6) described how he had solved a problem that "the very brilliant mathematicians who have studied the Theory of Errors" had overlooked and worked against the indifference of "the leading authorities in English statistics." Fisher (1939, p. 5) did acknowledge, however, that

It is doubtful if "Student" ever realized the full extent of his contribution to the Theory of Errors. From correspondence with him before the War...I should form a confident judgement that at that time he certainly did not see how big a thing he had done.

The same could be said of Fisher at the same time, but even when *Statistical Methods for Research Workers* appeared, Student did not realize how big a thing he was part of. He (1926, p. 148) welcomed the book that would change statistics as the first book to present the "special technique" required for dealing with small samples.

Had Student read his scientific obituary, he may well have said "oh, that's nothing—Fisher would have discovered it all anyway." That, according to Cunliffe (1976, p. 4), was his response on being thanked rather grandly for all he had done for "the advancement of statistics." Whether there was such an encounter, the tale is true to Student's dislike of pomposity, his modesty, his recognition of Fisher's genius, and his ease with it. The tale also brings out his realism, for Fisher would probably have discovered it all! From the start, Fisher was extraordinarily self-propelled and it is easy to argue that Student's intellectual impact was not on Fisher, but rather on Egon Pearson, who had doubts and discussed them with Student (see Pearson 1990, chap. 6).

The 1912–1925 interactions appear so sunny because there was a dark cloud in the form of Karl Pearson. Fisher saw his own situation as "publish or perish," and every Pearson rejection threatened his existence. He saw Student as a fellow victim, although Student was philosophical about Pearson, and in truth, Fisher could find no greater offense against him than "weighty apathy" toward his writing. Alas, new clouds were visible even in Fisher's memorial to his friend. When Fisher (1939, p. 6) suggested that concern with the "practical interpretation of experimental results" was vital for Student's success, he was taking a swipe not at Laplace or Gauss, but rather at Neyman and the younger Pearson. It became a theme with him that the latecomers had misunderstood his and Student's work—but that's another, and less sunny, story.

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Comment

A. W. F. Edwards

Zabell rightly stresses Student's pathbreaking contribution to statistical thought and practice with his 1908 paper, but let us not forget its literary quality too. The Introduction is a wonderfully clear description of the problem to be solved, the reason why it is important to solve it, and the means by which the author proposes to do so. It is a model of how to begin a scientific paper. The Conclusions at the end are equally clearly stated, and we may note particularly 'Finally I should like to express my thanks to Prof. Karl Pearson, without whose constant advice and criticism this paper could not have been written'.

For the fourth edition of *Statistical Methods for Research Workers* (1932) Fisher added a 'Historical Note' to Chapter I in which he said "'Student's' work was not quickly appreciated, and from the first edition it has been one of the chief purposes of this book to make better known the effect of his researches, and of mathematical work consequent upon them". Incidentally, in 1924 Fisher asked Student to read the proofs of the first edition, and one consequence of this was the incorporation of Student's suggestion that fold-out duplicates of the statistical tables in the book should be added at the end (Edwards, 2005).

It is a mark of the completeness of the revolution in statistical thinking which Student brought about that so little more needs to be said, but Zabell's account of how the mathematical gaps in his argument were later filled, the proofs improved, and the antecedents unearthed, is most welcome. Just one nagging problem remains – fiducial inference, to which Zabell turns in Section 4.4, having already mentioned a 'particularly interesting remark' of Student's in Section 3.2.

Both Zabell and Fisher (1939) have noticed that Student wrote 'if two observations have been made and we have no other information, it is an even chance that the mean of the (normal) population will lie between them', and Fisher remarked that this was an example of a statement of fiducial probability. He went on to note that it could be applied to the median of any distribution, and he generalised the method to samples of any

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size. Since Fisher wrote, we have become much more cautious about the interpretation of such statements, and I do not think they are fiducial either in the sense in which Fisher originally meant or in the sense in which most modern commentators use the word. By contrast, they are confidence statements.

When Neyman (1934) introduced confidence statements Fisher, in the discussion to Neyman's paper, immediately objected that unless they were based on sufficient statistics they were not using all the available information, and that conflicting confidence statements might be constructed on the same data. The discussion is unaccountably omitted from his Collected Papers, but he said:

Naturally, no rigorously demonstrable statements, such as these are, can fail to be true. They can, however, only convey the truth to those who apprehend their exact meaning; in the case of fiducial statements based on inefficient estimates this meaning must include a specification of the process of estimation employed. But the process is known to omit, or suppress, part of the information supplied by the sample. The statements based on inefficient estimates are true, therefore, so long as they are understood not to be the whole truth. Statements based on sufficient estimates are free from this drawback, and may claim a unique validity. (Fisher, in a footnote to his discussion of Neyman, 1934, 617–8.)

(His use of 'fiducial' had not yet settled down, and here of course refers to any statement with the confidence property, as we should now say.) Fisher (1934) essentially extended the field of application of fiducial probability to include location-and-scale distributions, showing how one must condition on the configuration of the sample.

Turning now to Student's statement, if $|x_1 - x_2|$ is very large, surely the probability that the mean lies between x_1 and x_2 is more than one half? Ought we not therefore to condition on $|x_1 - x_2|$ so that the reference set is that of all samples with the same 'width'? It is a simple matter to do this, and it is numerically convenient to use a normal distribution with unit variance as an example of the principle rather than the t-distribution. Table 1 shows the probabilities that the mean then lies between x_1 and x_2 for a selection of values of the width, taken from Table 23 of *Biometrika Tables Volume 1* (Pearson and Hartley, 1954).

Table 1.

Width $\lvert x_1 - x_2\rvert$	P(mean lies between x_1 and x_2 $\mid$ $\lvert x_1 - x_2\rvert$)
0.25	0.1403
0.5	0.2763
0.9539	**0.5**
1.0	0.5205
2.0	0.8427

Just as in the location-and-scale case, we have identified subsets indexed by the configuration of the sample, each with its own probability. Ignore them, and Student's 'confidence' statement holds. Are these then statements of fiducial probability? I think they are, because they are the same as the statements obtained from the straightforward fiducial distribution of a normal mean from a sample of two, so they respect the likelihood principle.

To put the matter plainly, suppose I play a game against Student and Fisher in which a sample of two is drawn from a normal distribution of unit variance, and we can then choose whether to bet that it embraces the mean, me paying up if it does not and they if it does. I can take money off them by declining to bet when the width is less than 0.9539 (see Table 1). How ironical to be able to beat Fisher by respecting his own original insistence that for a fiducial probability to be valid it must make use of all the information! But of course both men would have seen the point immediately and declined to play.

I note that Zabell helpfully gives references to the discovery that recognizable subsets similarly occur in the case of Student's t, but modestly omits a reference to an important paper of his own 'R.A.Fisher and the fiducial argument', Zabell (1992). (Or perhaps he has had second thoughts about its opening sentence 'Fiducial inference stands as R.A.Fisher's one great failure'!)

Some comment is needed on Zabell's suggestion that Aldrich (2000) and I (Edwards, 1995) are in 'dispute' over the role of E.J.Maskell in the birth of fiducial probability. I think these two sources are complementary. Aldrich's paper is a well-researched account of the similar ideas of M.Ezekiel, H.Hotelling, and H.Working in 1929–30; mine is an anecdotal mention of E.J.Maskell, whom Fisher credited with having had the idea whilst working with him at Rothamsted Experimental Station which Maskell had left in 1926. Hotelling spent some months at Rothamsted in the second half of 1929. The note I made in 1972 on which my 1995 anecdote is based reads "Norman Stenhouse (Adelaide) reports that Fisher told him he thought of and named the fiducial probability argument early in the 1920's when Maskell, at Rothamsted, said that you could imagine t 'confidence limits' not only for 1%, 5%, etc., but also all points in between. 'Let us call it the 'faith' or 'fiducial' distribution' ". See also Box (1978) for a slightly different slant. Perhaps this is no more than an example of an idea whose time had come. In my paper (1995) I wrote 'But it was Fisher who captured the idea as it floated past like a butterfly, examined it under his logical microscope, and announced it to the statistical world as being worthy of further investigation'. Nor should we forget the simultaneous developments by E.S.Pearson and J.Neyman, for which see David and Edwards (2001).

Mention of F.J.M.Stratton (Section 4.1) provides an opportunity to comment on the fact that he is often described as having been Fisher's 'tutor', which is not the case. Zabell (personal communication) has pointed out to me that the error stems from Student himself, when he so described Stratton in a letter to Karl Pearson dated 12th September 1912 (E.S.Pearson, 1968). It is a trivial matter of the difference between Oxford and Cambridge terminology. The fact that several Cambridge-trained authors have repeated it must be due to their having considered it unworthy of comment. In Oxford, Student's university, a tutor is indeed a college Fellow who teaches an undergraduate. The Cambridge equivalent is 'supervisor'. By contrast, a Cambridge tutor is responsible for a student's general well-being and will only coincidentally be in the same subject area. Fisher's tutor was W.W.Buckland, distinguished in Roman law and soon (in 1914) to become Regius Professor of Civil Law. Stratton was a College Lecturer in Mathematics, and in that capacity will have been teaching Fisher as a supervisor.

Finally, there are one or two minor points to note. Oxford (and Cambridge) degrees are not 'classed' (Section 2) – only the individual examinations are – and Karl Pearson really did spell 'antedates' 'antidates' (Section 3.3.3) as Zabell records! Last, and surely least, in Student's paper itself there are some arithmetical errors in the bottom line of the table on page 22.

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Comment

Eugene SENETA

1. NORMAL VARIANCE MIXING

The distributional aspect of Student's result is that $T = Z/\sqrt{V/m} \sim t_m$, where m is a positive integer, $Z \sim \mathcal{N}(0, 1)$, and $V \sim \chi^2_m$, with Z and V independently distributed. Adding to its richness of application, the Student statistic T may also be viewed (providing $m > 2$) as

$$T \stackrel{d}{=} \sigma W^{1/2} Z, \qquad EW = 1, \tag{1}$$

where Z is as earlier, $\sigma\,(> 0)$ is a constant, and W is a positive nondegenerate random variable distributed independently of Z. It is this representation, and the heavy tails of the t-distribution, that have led to its resurgence of returns in financial modeling in recent decades (see, e.g., Praetz 1972; Madan and Seneta 1990; Heyde and Kou 2004).

A context for the resurgence is a recent general model in financial theory that gives the price P_t of a risky asset over time $t \geq 0$ as

$$P_t = P_0 \exp\{ct + \theta T_t + \sigma B(T_t)\}, \tag{2}$$

where c, θ, and $\sigma\,(> 0)$ are real constants. Here $\{T_t\}$, the (market) activity time, is a positive increasing random process with stationary differences $W_t = T_t - T_{t-1}$, $t \geq 1$, that is independent of the standard Brownian motion $\{B(t)\}$. The corresponding returns (log increments) at unit time intervals are then given by

$$\begin{aligned} X_t &= \log P_t - \log P_{t-1} \\ &= c + \theta(T_t - T_{t-1}) + \sigma\big(B(T_t) - B(T_{t-1})\big). \end{aligned} \tag{3}$$

We assume that $EW_t < \infty$, and thus, without loss of generality, that

$$EW_t = 1, \tag{4}$$

to make the expected activity time change over unit calendar time equal to one unit, the scaling change in time being absorbed into θ and σ, while noting that

$$\begin{aligned} \sigma\big(B(T_t) - B(T_{t-1})\big) &\stackrel{d}{=} \sigma(T_t - T_{t-1})^{1/2} B(1) \\ &= \sigma W_t^{1/2} B(1), \end{aligned} \tag{5}$$

which is of form (1).

The case where $T_t = t$, $\theta = 0$ of model (2) is the classical geometric Brownian motion (GBM) model for the process $\{P_t\}$, with corresponding returns being iid $\mathcal{N}(\mu, \sigma^2)$. It goes back to Bachelier (1900) and is the basis of the now-famous Merton–Black–Scholes option pricing theory (see, e.g., Hull 2003, chap. 12).

Early investigations of data revealed that the distribution of returns, while retaining symmetry about the origin, gave Pearsonian kurtosis well in excess of 3, suggesting that the modeling of returns should be by a symmetric distribution with heavier tails than the normal. In particular, Praetz (1972) argued in favor of variance dilation of the normal through variance mixing and found that mixing according to $X|W \sim \mathcal{N}(\mu, \sigma^2 W)$, where W has an inverse-gamma ($I\Gamma$) probability density function (pdf)

$$f_W(w) = \begin{cases} \dfrac{(\nu/2-1)^{\nu/2} w^{-\nu/2-1} e^{-(\nu/2-1)/w}}{\Gamma(\nu/2)} \\ \quad \text{if } w > 0 \\ 0 \quad \text{otherwise,} \end{cases} \tag{6}$$

where $EW = 1$ as required, providing that $\nu > 2$, gives for the pdf of returns the scaled t-distribution symmetric about 0 and with degrees of freedom ν (not necessarily an integer) with pdf

$$f_X(x; \nu, \delta) = \frac{\Gamma((\nu+1)/2)}{\delta\sqrt{\pi}\,\Gamma(\nu/2)} \cdot \frac{1}{(1 + ((x-\mu)/\delta)^2)^{(\nu+1)/2}} \tag{7}$$

with $-\infty < x < \infty$, where $\delta = \sigma(\nu-2)^{1/2}$. The classical (Student's t-distribution) form has $\delta = \sqrt{\nu}$, with ν a positive integer.

Influenced by Praetz's article, Madan and Seneta (1990) took the distribution of the mixing variable W itself to be Γ (gamma,

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rather than inverse gamma) in the case where $\theta = 0$. This led to a continuous-time model now known as the (symmetric) Variance-Gamma model. In the general subordinator model, the distribution of returns is given by

$$X|W \sim \mathcal{N}(\mu + \theta W, \sigma^2 W). \tag{8}$$

Incidentally, the choice of Γ or the $I\Gamma$ distribution for W leads to skewed versions of variance-gamma and t distributions, and gives one method of producing a skew version of Student's t-distribution.

However, the symmetric scaled t-distribution with pdf (7) continues to enjoy favor because of its power law (Pareto-type) probability tails, a property manifested in the nonexistence of higher moments. Empirical investigations initiated by Heyde (1999) tend to favor nonexistence of higher moments for a model of returns. So this too is a legacy of Student's first 1908 article and continues the empirical spirit in the use of real (financial) data sets of both articles published that year.

2. BAYESIAN SUBTEXT

A theme that runs through Sandy Zabell's meticulously researched account is that of Bayesian context, not least in Student's thinking. Particularly interesting is the remark to Fisher on April 3, 1922, that the only prior that Student found acceptable around the time that the two 1908 articles were prepared was the uniform, which was not only consistent with Fisher's own maximum likelihood method of estimation, but enabled Student to handle the mathematics (Pearson 1990, pp. 28–29). The Bayesian history of the t-distribution of which Student was unaware, and which Sandy Zabell mentions, specifies a joint improper uniform prior for location μ and precision H, and conditional distribution of observation X as $X|(\mu, H) \sim \mathcal{N}(\mu, (2H)^{-2})$. On the basis of observed sample values $x_1, x_2, \dots, x_n$ the marginal posterior distribution for μ is then (7) with x replaced by μ, μ replaced by $\bar{x}$, with $\nu = n$ and $\delta^2 = (n-1)s^2/n$, where $\bar{x}$ and s are the usual sample mean and sample standard deviation.

I wonder whether Student may have dabbled with the following simpler Bayesian steps. If the sample is taken from the normal, with $X|\Theta \sim \mathcal{N}(0, \Theta)$, where Θ has an improper uniform prior over $(0, \infty)$, the posterior pdf of Θ is given by an $I\Gamma$ distribution. We have seen that by mixing on the variance of a normal by an inverse gamma, and in this setting the mixing distribution taken as the posterior distribution for the normal variance, we obtain a t-distribution for the predicted random variable X. At the heart of both arguments is the Γ pdf, essentially a Pearson type III curve.

3. RUSSIAN CONNECTIONS

Student's methodological approach through moments to guessing that under normal sampling of size n, $(n-1)S^2/(\sigma^2) \sim \chi^2_{n-1}$ on the basis of third and fourth moments, and his approach to the independence of $\bar{X}$ and S^2 by establishing uncorrelatedness again using moments, remained an acceptable (although of course intrinsically flawed) means of justification for many years. We illustrate how Markov and Chuprov were led by moment arguments to conjecture a limiting Pearson type III distribution, when, like Student, they considered the distribution of a ratio of random variables.

At the time that Student was working on the two 1908 articles, the ideas of Karl Pearson's English Biometric School were gaining currency among the statisticians of the Russian Empire. In addition to articles and books by L. K. Lakhtin (1903), A. A. Chuprov (1910), A. V. Leontovich (3 volumes in 1909–1911), and R. M. Orzhentskii (1910), there was the book of Evgenii Evgenievich Slutsky (1912), whose name was to become famous in econometrics as well as in mathematical statistics through his study of stochastically dependent random processes. His name now attaches to Slutsky's theorem, and the Slutsky effect in time-series analysis, for example.

Slutsky had spent a full year studying the publications of Karl Pearson and his disciples. His book was a synthesis of that work by a mathematically well-educated political economist, as he then was. The book was in fact Slutsky's very first publication. Its appearance helped him gain a position at the Kiev Commercial Institute in Spring 1915, where he worked, rising to the rank of full professor, until 1926, when he left for Moscow. The book comprises two parts, as its title indicates. The second (and main) part, on the theory of correlation, exposits the analysis of "probable errors" of estimators associated with bivariate samples, along Pearsonian lines. Under this theory, which is in effect large sample–based with underlying approximate normality of estimators, the probable error of the sample correlation coefficient r is given by

$$E_r = .67449 \frac{1 - r^2}{\sqrt{N}}, \tag{9}$$

where N is the sample size (I use Slutsky's notation). The value .67449 is the standard multiplier of what is effectively the standard deviation of the estimator in Pearsonian theory as Slutsky presents it; it is the 75th percentile of the standard normal distribution. In his account, it is Student's second 1908 article that is of interest to him. He studied it meticulously and cited its main conclusions, writing that:

> According to the investigations of "Student," formula [*formula (9) here*]... may be applied at values of N no less than 30. For smaller samples it cannot be relied on, and another method of assessment must be applied.... As a practical rule, for N between 20 and 30 the sample correlation coefficient should be no less than 0.5 [*in absolute value*] in order to speak of existence of nonzero [*population*] correlation with some degree of certainty.... In conclusion we must admit with regret that on the basis of tables constructed according to "Student" 's formula, the statistician should not apply correlational theory at all to samples of size less than 20 (pp. 97–98).

The Markov and Chuprov correspondence (see Ondar 1981; Sheynin 1996 for supplementation), which began in 1910, was later (around 1916–1917) affected by the Pearsonian theory, as exposited in the first part of Slutsky's book.

Apart from the normal, the curves of the Pearson family fall into six types. The type of a Pearson curve is determined by the value of a "criterion constant" k,

$$k = \frac{\beta_1(\beta_2 + 3)^2}{4(4\beta_2 - 3\beta_1)(2\beta_2 - 3\beta_1 - 6)}, \tag{10}$$

where

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}, \qquad \beta_2 = \frac{\mu_4}{\mu_2^2}, \tag{11}$$

with the usual meaning for the μ_i.

Markov and Chuprov were concerned with a variant of Lexis' ratio,

$$Q^2 = \frac{\sum_{i=1}^{m}(P_i - P)^2/m}{P(1-P)/n}. \tag{12}$$

Here a block of n trials, at each of which success or failure can occur, is replicated independently m times. P_i is the proportion of successes observed in the ith block, and P is the proportion of successes in all mn trials.

If the trials within each block are independent and the probability of success at each of the mn trails is constant, then Chuprov proved

$$EL = 1, \quad \text{where } L = \{(mn-1)/n(m-1)\}Q^2, \tag{13}$$

provided that L is defined as unity when $P(1-P) = 0$ (see Ondar 1981, letter 80, pp. 94–100).

Chuprov, considering $E(Q^6)$ and $E(Q^8)$ concluded that the limiting distribution of Q^2 would be skewed (Sheynin 1996, chap. 8, letter 88a). Markov pushed him (January 28, 1917; Ondar 1981, letter 89, pp. 114–115) to describe the limit in terms of a Pearsonian curve. Chuprov responded (January 28, 1917; Sheynin 1996, chap. 8, letter 88c) that the Pearsonian quantity

$$2\beta_2 - 3\beta_1 - 6 \tag{14}$$

is "precisely" 0, suggesting a type III curve.

Markov then sketched, using moment arguments (Ondar 1981, letters 93 and 94, pp. 118–122), the limiting distribution as $n \to \infty$ of

$$\{(mn-1)/n\}Q^2, \tag{15}$$

which leads to a χ^2_{m-1} distribution, an instance of a type III curve, and hence of a Γ pdf.

This result has a resemblance to and connection with the distribution of the variance ratio in one-way ANOVA, where under normal theory and the null hypothesis, the numerator and denominator variances are independently distributed chi-squared statistics (Heyde and Seneta 1977, sec. 3.4). Thus there is a somewhat remote connection with the distribution of the square of the Student statistic.

Starting from the correspondence, Chuprov was led to study quantities such as $E(\frac{X}{Y})$ where X and Y are not necessarily independent random variables. But his primary focus subsequently became correlation theory, presumably under Pearsonian influence and the influence of Slutsky's (1912) book. He did, however, cite Student's second article in the notes to chapter 6 of his book, along with Fisher's follow-up article on a "normal correlation" to that. (The English translation of Chuprov's book is Tschuprow 1939.)

There was correspondence between Karl Pearson and Chuprov's student Oskar N. Anderson, then in his 20s, in early 1914 partly relating to Student's 1914 article (Student 1914) on the variate-difference method. Sheynin (1996, secs. 15.6 and 17) reproduced Karl Pearson's response to Anderson, in which the following passage occurs (the English translation is mine):

> Student ist nicht ein Fachmann, doch glaub'ich dass Sie zu vieles Gewicht an seine Wörter liegen, wenn Sie sagen dass er glaubt... [Student is not a professional, and I believe that you attach too much significance to his words, when you say that he believes...].

This passage continues on to justify Student. Karl Pearson's evaluation of Student in this context seems less surprising than the statement of Student to Egon Pearson about Fisher's inexperience, which Zabell quotes at the end of Section 4.3.1.

In any case, the correspondence between Karl Pearson and Oskar Anderson (Sheynin 1996, secs. 15.6 and 17) is another indication of awareness of Student's work by statistical scholars in the Russian Empire. Perhaps it is true, as Student lamented in a letter to Fisher in 1934, that in pre-Fisher days no one paid the slightest attention to the first 1908 article. People in the emerging Statistical School of the Russian Empire did, however, pay attention to the second 1908 article as early as 1912, and soon after to the other papers as well.

Anderson moved to Kiev and in 1917 began to lecture in mathematical statistics at the Kiev Commercial Institute, where he and Slutsky would have had much in common, particularly a shared interest in statistical dependence. However, he was there a short time only before becoming an émigré in dramatic circumstances, ultimately achieving great eminence in Germany and in the West.

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Comment

Persi DIACONIS and Erich LEHMANN

Sandy Zabell gives us a good feel for Student and his times. We supplement this with later developments showing that Student's intuitive test is optimal in various senses (with some serious caveats). We also show how Fisher's and Hotelling's geometric interpretation of Student's test percolates to a wide variety of statistical applications. These developments bring the historical record up to the present.

1. SOME HISTORY

Student's discovery of the t-distribution did not come out of a vacuum. As he wrote in the introduction to his 1908 article:

> The usual method of determining the probability that the mean of the population lies within a given distance of the mean of the sample, is to assume a normal distribution about the mean of the sample with a standard deviation equal to $S/\sqrt{n}$ where S is the standard deviation of the sample, and to use the tables of the probability integral (i.e., the standard normal distribution).

A few paragraphs later, he added:

> Although it is well known that the method of using the normal curve is only trustworthy when the sample is "large", no one has yet told us very clearly where the limit between "large" and "small" samples is to be drawn.

The use of the t-statistic for the problem at hand was thus in consistent use in Student's time. A detailed discussion of Laplace's work on its distribution was provided by Hald (1998, sec. 20.8), who raised the question why Laplace did not find the t-distribution, with the answer that "he fell for the easy normal approximation."

2. OPTIMALITY

The use of the sample mean $\bar{x}$ to estimate the population mean μ, with division by the estimated standard deviation for scaling purposes, seems intuitively so clearly the right thing to do (assuming that the sample comes from a normal distribution) that one might expect the one-sided t-test to be uniformly most powerful (UMP), that is, for no improvement to be possible.

Unfortunately, however, this is not the case. It turns out (Lehmann and Stein 1948) that at levels $< \frac{1}{2}$, it is possible to improve on the power of the t-test against any particular alternative, the improvement being tailored to that alternative (with a relative loss of power against other alternatives). But the situation changes when the condition of unbiasedness is imposed. Both the one-sided and two-sided t-tests are UMP among all unbiased tests (Neyman and Pearson 1936a,b, 1938).

Another optimality property is defined by noting that both the hypothesis $H:\mu = 0$ and the class of alternatives $\mu > 0$ remain invariant (i.e., unchanged) if all of the observations are multiplied by a common positive constant c, because then both μ and the standard deviation σ are also multiplied by c. It is then natural to restrict attention to tests that are invariant under the group G of all such scale changes. Among all such invariant tests, the t-test is UMP. (The corresponding result holds for the two-sided t-test by allowing both positive and negative c.)

The fact that the t-test is UMP-invariant implies another, very different optimality result. The group G is an amenable group (a class of groups containing compact or abelian groups and their extensions); thus it follows from the Hunt–Stein theorem that the one-sided t-test maximizes the minimum power over the alternatives $\mu/\sigma = k$ for any positive k, and thus also, for example, over the alternatives $0 < \mu/\sigma < k$ or $\mu/\sigma > k$. For the two-sided case, μ/σ must be replaced by $|\mu|/\sigma$. (See Lehmann and Romano 2005, sec. 8.5, for more on the Hunt–Stein theorem.)

The foregoing optimality results do not require knowledge of the power function. But such knowledge is needed if one wants to determine the sample size that achieves a given power. This power is given by the noncentral t-distribution with $n-1$ degrees of freedom and noncentrality parameter $\sqrt{n}\mu/\sigma$. The family of noncentral t-distributions for varying noncentrality parameters has a monotone likelihood ratio, from which it follows that the t-test is UMP-invariant for testing $\mu/\sigma \le c$ against $\mu/\sigma > c$ for any $c > 0$. Further optimality properties (e.g., admissibility) have been discussed by Anderson (2004, sec. 5.6).

3. TWO CAVEATS

The robustness properties that Zabell discusses, together with the optimality properties mentioned in the preceding section, present the t-test as a very desirable solution to the testing problem in question. Both must be taken with a grain of salt, however.

Consider first the test's robustness against nonnormality. Evidence for this is provided by some empirical studies by E. S. Pearson (and later by a very extensive and careful investigation by Posten [1979]). A different type of robustness occurs because the distribution of the t-statistic is asymptotically normal for all distributions that lie in the domain of attraction of the normal distribution. For those distributions, therefore, the asymptotic level of the t-test is equal to the nominal level.

Note, however, that Bahadur and Savage (1956) showed that for any fixed sample size, no matter how large, there exist distributions with finite variance for which the level of the t-test is arbitrarily close to 1. Of course, this does not contradict the pointwise convergence to the nominal level, but only shows that this convergence is not uniform. (Uniform and nonuniform convergence of the level of the t-test over various classes of distributions was investigated further in Romano 2004.) These results suggest that a certain amount of caution regarding robustness is required for finite sample sizes.

Let us now turn to the optimality results of Section 2. These hold under the assumption that the underlying distribution is

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normal, but even under slight deviations from normality, the t-test can be far from optimal. The poor performance of the t-test, particularly for distributions with heavy tails, can be seen in comparison with nonparametric tests, such as the Wilcoxon or normal scores tests. The asymptotic relative efficiency of t to either of those tests can be arbitrarily close to zero for sufficiently heavy-tailed distributions with finite variance. On the other hand, for all distributions with finite variance, the asymptotic efficiency relative to t is $\geq .864$ for Wilcoxon and ≥ 1 for normal scores (Hodges and Lehmann 1956; Chernoff and Savage 1958). Of course, there are other possible breakdowns; for a useful discussion of how dependence affects things (see Miller 1997, chap. 1).

4. THE COST OF NOT KNOWING σ

To test the mean of a normal distribution with known σ, one would divide the sample mean $\bar{x}$ by σ rather than by its estimate S. The resulting test would then be more powerful than the t-test. To see how much is lost by using t instead of $\bar{x}/\sigma$, one can note that the asymptotic relative efficiency of the latter test to t is 1, which suggests that the loss is small.

More detailed information can be obtained by computing deficiency, that is, the limit (as the sample size $\to \infty$) of the difference between the number of observations required by the two tests to achieve the same forms (Walsh 1949; Hodges and Lehmann 1970). This limit depends on α and, for $.01 \leq \alpha \leq .1$, lies between .8 and 2.7. Thus the number of "wasted observations" is quite small even for large samples.

5. THE PERMUTATION t-TEST

As Zabell discusses in Section 4.7.2, Fisher proposed a permutation version of the t-test that has an exact level without the assumption of normality, provided that the underlying distribution is symmetric, or even without any distributional assumptions if the data are obtained through suitable randomization.

Fisher conjectured that the t-test and its permutation version are nearly equivalent, and confirmed this in an example. The two tests differ only in the critical value for the t-statistic, this value being constant for the t-test and random for its permutation alternative. Hoeffding (1952) showed that these two values tend to the same limit, and that asymptotically the two tests have the same power. These results justify Fisher's suggestion that the usual t-test can be viewed as an approximation to its distribution-free permutation version.

Because the t-distribution tends to the normal as the sample size n tends to ∞, a third asymptotically equivalent test consists in using the normal critical value. The problem of when the normal or t critical value provides the better approximation was solved by Diaconis and Holmes (1994), who also provide algorithms for the exact evaluation of the permutation distribution.

6. SOME EXTENSIONS

So far, we have reviewed various properties of the t-test. For the remainder of these comments, we consider some of the many extensions and applications of this test that have made Student's work so influential.

In his 1908 article, Student considered only the paired-comparisons (or, equivalently, the one-sample) problem. In 1922, Fisher pointed out that the t-distribution also provided the solution to the two-sample problem (with equal variances) and that of testing a regression coefficient. A crucial further extension (Fisher 1924) was to the F-distribution, which allowed the testing of several means or regression coefficients or, more generally, testing the general univariate linear hypothesis.

The t-test was generalized further to the testing of a multivariate normal mean vector (Hotelling 1931) and later also to the sequential case (Wald 1947). In the remainder of this comment, we consider in more detail still another generalization, which is less well known.

7. GEOMETRIC UNDERSTANDING AND THE VOLUME OF TUBES

Fisher's (1915) geometric proof of the distribution of the t-statistic has been extended by Hotelling, Efron, Eaton, and others. It leads to a substantial part of modern differential geometry (Weyl's volume of tubes formulas [Weyl 1939]).

Suppose that $X_1, X_2, \ldots, X_n$ are independent with a normal (μ, σ^2) distribution. The two-sided t-test for $\mu = 0$ rejects if $|T|$ is large, where

$$|T| = \frac{\sqrt{n}|\bar{X}|}{S_n}, \qquad \bar{X} = \frac{1}{n}\sum_{1}^{n} X_n,$$

and

$$S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Fisher came to a geometric understanding of both the statistic and its null distribution as follows. Consider the vector $X = (X_1, \ldots, X_n)$. If $\mu \neq 0$, then the vector will be close to the vector $(\mu, \mu, \ldots, \mu) = \mu\mathbf{1}$, with $\mathbf{1} = (1, 1, \ldots, 1)$. Thus the angle between X and $\mathbf{1}$ will tend to be small. On the other hand, if $\mu = 0$, then X, projected onto the unit sphere, is uniformly distributed. Thus a simple, sensible test of $\mu = 0$ is as follows: Project X onto the unit sphere. Look at the distance to the vector $(\frac{1}{\sqrt{n}}, \ldots, \frac{1}{\sqrt{n}})$. Reject if this distance is small. To calibrate this test, take a cap around $(\frac{1}{\sqrt{n}}, \ldots, \frac{1}{\sqrt{n}})$ with (spherical) area .05% of the total area of the unit sphere. The test rejects if the projection of X is in the cap. Fisher showed that this is precisely the t-test with the cap area the null distribution of the t-statistic.

Hotelling (1939) considered paired data $(X_1, Y_1), \ldots, (X_n, Y_n)$ modeled as

$$Y_i = ae^{bX_i} + \epsilon_i,$$

with ϵ_i independent $n(0, \sigma^2)$ errors. To test $a = 0$, Hotelling suggested the following geometric test. Consider the vector $v_b = (e^{bX_1}, \ldots, e^{bX_n})$. As b varies, this vector, projected onto the n-dimensional unit sphere, spans a curve. If $a = 0$, then the projection of Y onto the sphere is uniformly distributed. Thus one may test $a = 0$ by seeing if the projection of Y is unusually close to the curve. For this, form a tube around the curve with area 5% of the total and reject by assessing whether the projection of Y falls in this tube.

Thus Hotelling needed to find the volume of a tube around the curve. Consider (along with Hotelling) a curve in the plane and a tube of small radius r around this. If the curve is a straight line, then of course the volume is $2r \times$ length. Now consider bending the line. Infinitesimally, the volume of a wedge of one side of the line gets bigger and that on the opposite side of the line gets smaller. Hotelling showed that there is exact cancellation! This result is not only for r infinitesimally small. As long as there is no global self-intersection, the volume of a tube around a curve is $2r \times$ length. The analogous result holds for curves in high dimensions, as well as for curves on the sphere. (See Johansen and Johnstone 1990 and Johnstone and Siegmund 1989 for background, details and further applications of Weyl's formula to statistics.)

To carry out such tests for two-parameter models [e.g., Fisher's test for periodicity where $y_i = a + b\cos(cx_i + d) + \epsilon_i$], Hotelling needed to solve the higher-dimensional version of the problem of determining the volume of a tube around a surface. Of course, if the surface is a part of a flat plane in $\mathbb{R}^n$, then the volume is

$$2r \times \text{surface area.}$$

If a flat surface is bent, say upward along parallel lines, then this result continues to hold by the one-dimensional theorem. It is difficult to picture what happens if the surface is twisted in space. Hotelling could not solve this problem but gave a talk at the Princeton Mathematics Club on the subject in the late 1930s. Herman Weyl was in the audience, and he solved the problem. The volume of a tube of radius r about a surface is

$$2r \times \text{ surface area } + cr^3 \times \text{average Gaussian curvature,}$$

with c a universal constant. He found similar formulas for the volume of a tube around k-dimensional surfaces inside constant curvature spaces such as $\mathbb{R}^n$ or the n-sphere. In all cases, the volume is a polynomial of degree $1 + [\frac{k}{2}]$ in r with intrinsically defined coefficients (e.g., the coefficients do not depend on how the surface is embedded).

Weyl's "higher-order intrinsic curvatures" were new in geometry. They led Chern to his development of the modern version of the Gauss–Bonnet theorem and much else. The story was well told by Osserman (1978) and Gray (2004). It has had application in statistics, particularly to the original problem of Hotelling, for periodicity in a time series (see Knowles and Siegmund 1989). (For confidence sets in nonlinear regression, see Naiman 1990; for applications to projection pursuit, see Sun 1993; for the maximum of Gaussian processes, see Adler and Taylor 2007.)

8. FROM VOLUME TESTS TO ALGEBRAIC STATISTICS

Some of the appeal and robustness properties of volume tests come from their geometric description (see Efron 1969; Eaton and Efron 1970; the discussion in Eaton 1989, pp. 63–67). This led Diaconis and Efron (1985) to suggest volume tests in a variety of other problems. For example, consider tests of independence for an $I \times J$ contingency table. The surface of independence sits in the $(IJ - 1)$-dimensional simplex S of all probabilities on IJ cells. The observed table can be mapped into S (by dividing N_{ij} by the total N). A test of independence rejects if the observed table is close to the surface. This requires a tube around the surface of volume .5% of the total volume of S. Computing this volume is difficult, as is the justification of uniform measure on S. A second volume test involves conditioning on the row and column sums and sampling tables uniformly with these sums fixed. This algorithmic task has lead to a healthy development that brings techniques of algebraic geometry into the picture.

Consider the task of generating a random $I \times J$ table with fixed margins $N_{i\cdot}, N_{\cdot j}$, $1 \le i \le I, 1 \le j \le J$. A Markov chain Monte Carlo algorithm proceeds as follows. From one table T satisfying the contraints, pick a pair of rows and a pair of columns randomly. Change the entries of T in the four determined cells by adding and subtracting one in the pattern (choose with probability 1/2)

$$\begin{matrix} + & - \\ - & + \end{matrix} \quad \text{or} \quad \begin{matrix} - & + \\ + & - \end{matrix}$$

If this results in a table with nonnegative entries, then the change is made; otherwise, the walk stays at T. It is not hard to see that this walk gives a connected Markov chain with a uniform stationary distribution on the set of tables with the fixed row and column sums.

Finding similar walks for higher-way tables or related problems in logistic regression defeated all comers. Diaconis and Sturmfels (1998) managed to cast this as a problem in computational algebra. The computer successfully finds such moves which are in routine use, through such programs as Latte (lattice point enumeration; available at *www.ucdavis.edu/~latte/*). The algebraic geometric techniques then found application in design (Pistone, Riccomagno, and Wynn 2001) and in determining the behavior of complex singularities for maximum likelihood in log-linear models (Hosten and Meek 2006).

Although far from the source, there is a clear trail back through Hotelling's volume test, Fisher's geometrical view, and thus back to Student's original 1908 article.

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Rejoinder

S. L. Zabell

I thank all of the discussants for their varied and stimulating contributions. I will confine my response to some common themes and matters of detail.

Professor Stigler questions whether Student was an essential catalyst in Fisher's statistical development; Professor Aldrich notes, in contrast, Egon Pearson's suggestion that Student may have played "some role" in Fisher's being hired at Rothamsted. The latter is an intriging possibility, but the available evidence suggests otherwise.

Sir E. John Russell, the director of Rothamsted from 1912 to 1943, has described the circumstances under which Fisher came to Rothamsted (Russell 1966, pp. 325–326):

> Dr Horace Brown...introduced me to Ronald Aylmer Fisher, a former mathematical master at one of the Public Schools who wished to change over from teaching to research work in Statistics.... From 1915 to 1919 [Fisher] taught mathematics and physics at various public schools; he was constitutionally unfitted for the work and was neither happy nor successful at it: when I first saw him in 1919 he was out of a job. Before deciding anything I wrote to his tutor at Caius College [that is, Stratton] whom I knew personally, asking about his mathematical ability. The answer was that he could have become a first class mathematician had he 'stuck to the ropes', but he would not. That looked like the type of man we wanted, so I invited him to join us.... It took me a very short time to realize that he was more than a man of great ability: he was in fact a genius who must be retained.

Joan Fisher Box's account in her biography of her father provides some additional detail (Box 1978, p. 61):

> On one of his visits to Cambridge, Fisher met Horace Brown, the botanist, to whom he had been introduced by Major Darwin during the war and for whom he had done some work.... [Brown] introduced him to Dr. E. J. Russell, who was visiting Cambridge looking for staff for Rothamsted Experimental Station. In August there followed an offer from Rothamsted....

It is in the crucial period 1922–1925 that, as Aldrich ably recounts, Student came to play a such an important role in Fisher's professional life. It is surely significant that after his 1915 article on the correlation coefficient, 7 years elapsed before Fisher began to study the small-sample distributions of other statistics, and then only in direct response to the stimulus of Student's two letters to him in April 1922.

But the best witness is Fisher himself and the way he regarded Student, as evidenced by his comment in the fourth edition of SMRW (noted by both Aldrich and Edwards) that it "it has been one of the chief purposes of this book to make better known the effect of [Student's] researches," his attempt to gain Student's election to the Royal Society (Pearson 1990, pp. 118–119), his warm tribute to Student in his obituary notice (Fisher 1937), and even the unusual step of preparing summaries of his correspondence with Student (the summaries appearing at the beginning of McMullen 1970). It is clear Fisher had no doubt about the crucial role that Student had played in his life.

Professor Edwards begins his discussion of Student's article by very rightly noting that we should "not forget its literary quality as well." In this he is in very good company. Egon Pearson, in his appreciation of Student, concurs: "It is a paper to which I think all research students in statistics might well be directed, particularly before they attempt to put together their own first paper...[T]here is something in the arrangement and execution of the paper which will always repay study" (Pearson 1939, p. 221).

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The fiducial example that Professor Edwards discusses is very interesting, illustrating as it does that in some situations conditional confidence and fiducial probability can coalesce. But when Edwards refers to "the sense in which most modern commentators use the word [fiducial]," I am not sure that such a consensus meaning really exists. Being myself a (philosophical) Bayesian, of course in principle I would always condition on the ultimate configuration, the sample itself.

I am also grateful for Edwards's comments regarding both Maskell and Stratton. Because Fisher's mentor Stratton has once again made a cameo appearance, some further comments about this very interesting figure may be in order. Frederick John Marrian Stratton (1881–1960) was the quintessential Caius man. After entering Gonville and Caius as a student in October 1901, Stratton remained there for the rest of his life. Third Wrangler in Part I of the Mathematical Tripos in 1904 (no mean distinction—Sir Arthur Stanley Eddington was Senior Wrangler that year), Stratton was elected a Fellow of Caius in 1906, and then served in a succession of posts: College Lecturer in 1910, Tutor in 1919 (after serving with distinction in the war and receiving the DSO), Senior Tutor in 1921, Professor of Astrophysics and Director of the Solar Physics Laboratory in 1928, and Fellow of the Royal Society in 1947. His obituary in the *Biographical Memoirs of Fellows of the Royal Society* was written by James Chadwick (1961), the Nobel laureate who discovered the neutron.

How did this young Cambridge astronomer come to know the older Oxford-trained chemist working in Dublin on statistical problems in brewing? It happened in 1910, when, as Student later wrote Karl Pearson (Pearson 1968, p. 446):

> It was on this subject [of small samples] that I came to have dealings with Stratton, for in a paper setting up to teach agriculturalists how to experiment he had taken as an illustration a sample of 4! I heard about it, wrote to the man whom I supposed to be writing the paper with him and he forwarded my letter to the guilty pair. They sent me their papers to correct the day before the proofs were sent in and I mitigated some of it!

Turning to Eugene Seneta's discussion, Karl Pearson would no doubt have been pleased to learn that one of the curves in his four-parameter family has found use in financial modeling. I was particularly interested to see Seneta's remarks on Russian connections with the English school of statistics, a topic also touched on by Aldrich (2003).

The discussion by Persi Diaconis and Erich Lehmann reminds us just how far mathematical statistics has come over the last 100 years. But could it not be argued that we are now in some ways at a juncture very similar to the one a century ago? Student was a transitional figure, linking the "Mark I" statistics of Karl Pearson and the "Mark II" statistics of Fisher. Today our field is once again in the process of transformation, as such areas as bioinformatics, data mining, and financial statistics present new challenges in which "small" can mean "in the millions" and "efficiency" refers to speed in algorithmic implementation when confronted with massive data sets. Who will be the Students and Fishers of postmodern statistics?

Perhaps we should leave the last word to Fisher, in the concluding paragraph to his obituary of Student (Fisher 1939):

> I have purposely laid emphasis on one scientific achievement with which his name will always be associated. His life was one of fruitful scientific ideas, and his versatility extended beyond his interests in research. In spite of his many activities it is the "Student" of *"Student's" test of significance* who has won, and deserved to win, a unique place in the history of scientific method.

ADDITIONAL REFERENCES

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